Rho negation-map audit (2026-10-20)

Question. Does every Pollard rho we use as a reference walk on the classes $\{P, -P\}$ (the negation map, $A = 2$), and does each such walk deliver the $\sqrt{2}$ saving in counted operations?

Answer. Before this change, two walks lacked the map: `volcano-rho`'s native walk (crypto) and crypto-autoresearcher's index-calculus baseline `index_calculus/rho.py` (audit X02/X03). Both now walk on $\{R, -R\}$ by default, with the old walk kept behind `--negation off` and `--rho-walk plain`. The matched reference `ecbench rho.negation` already had the map, and it delivers the saving.

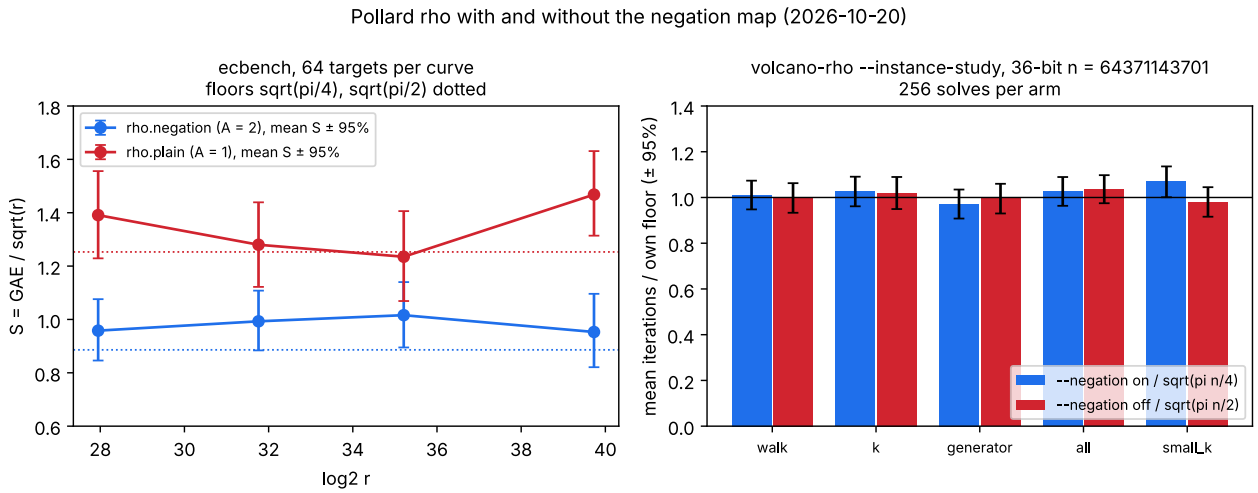


Figure 1: Left: ecbench session ECBS1h83b97eccdcda, mean S with 95% interval, 64 targets per curve and arm; dotted lines are the floors $\sqrt{\frac{\pi}{4}}$ and $\sqrt{\frac{\pi}{2}}$. Right: `volcano-rho --instance-study`, 36-bit prime-order curve, 256 solves per arm, mean iterations over each walk's own floor.

1. ecbench `rho.negation` vs `rho.plain`

Spec ECS1h7a0b49db98d3 at main `6e35e1f42` (clean). Four fresh prime-order curves (28–40 bits; $\#E$ counted natively, prime, cofactor 1). 512 records, all verified; 16 exact replays. Host: Intel Xeon @ 2.10 GHz, 4 vCPU shared VM, L0.

curve	log2 r	neg S [95%]	/floor	plain S [95%]	/floor	plain/neg
fp28-tm24531	27.95	0.958 [0.846, 1.076]	1.081	1.391 [1.229, 1.556]	1.109	1.452
fp32-t93403	31.76	0.993 [0.884, 1.108]	1.120	1.280 [1.122, 1.439]	1.021	1.289
fp36-t360541	35.21	1.016 [0.895, 1.140]	1.146	1.235 [1.069, 1.406]	0.986	1.216
fp40-tm62443	39.73	0.953 [0.821, 1.096]	1.075	1.468 [1.314, 1.631]	1.171	1.540

- Pooled ops (cluster bootstrap, 256 pairs): plain/negation = **1.371 [1.257, 1.491]**; $\sqrt{2}$ = 1.414 is inside. ecbench: "diagnostic only (unpriced work)".
- §5a band: the order holds at 0 and 0.1 GAE per canonicalisation. At 1 GAE the negation S is 1.89–2.03, above plain's. Verdict: unresolved inside the band on all four curves.
- Wall (L0, descriptive): median paired ratio 1.140.

2. `volcano-rho` (changed)

36-bit curve $p = 64371108127$, $n = 64371143701$; `--negation on` vs `off`, same seed, 256 solves per arm, 2,560/2,560 correct.

arm	on / $\sqrt{\pi n/4}$	off / $\sqrt{\pi n/2}$	off/on	ns/iter on	ns/iter off
walk	1.011	0.998	1.396	363	314
k	1.026	1.020	1.405	362	310
generator	0.971	0.995	1.449	362	306
all	1.027	1.036	1.428	336	308

small_k	1.068	0.980	1.298	353	314

Iterations include the look-ahead's discarded additions and the escapes' doublings. Cost per iteration rises 9–16%, so the wall gain is about 1.2× (descriptive). Native and pull-back lenses: 176/176 correct each on a 30-bit depth-2 volcano.

3. crypto-autoresearcher `index_calculus/rho.py` (changed)

640 paired solves at 16–32 bits: plain/negation walk_ops = 1.353 [1.281, 1.425]. The negation walk is at 1.19× its floor; plain at 1.14×. With `--rho-walk plain` the output is bit-identical to the walk before the change.

Scope

This corrects a reference and makes two rho baselines stronger. It is not an index-calculus improvement and moves no leaderboard row; crypto's prime rows already use `rho.negation`. Rows built on the autoresearcher's plain rho are $\sqrt{2}$ too generous to IC. Not changed: crypto-autoresearcher `harness/rho.py` and `harness/walk.py` (follow-up). Sources: `research/rho_negation_audit_20261020/` (crypto) and `research/rho_negation_map_20261020/` (crypto-autoresearcher).